

VD2 Session Transcript

October 5-7, 2026. User prompts and assistant responses, including progress updates, through the PDF request. Original wording and Markdown notation are preserved. System instructions, environment metadata, and tool logs are excluded.

1. User

C:\Users\bedna\OneDrive\Documents\AI-Class\VD2_student_files

here is the link to our files. python VD2_analysis.py compare --csv data/train.csv --out outputs do not evaluate

2. Assistant

I'll run the comparison command in that folder without running evaluation.

3. Assistant

The CSV is directly in the folder, so I'll use `--csv train.csv` to run the comparison.

4. Assistant

Comparison completed successfully using `--csv train.csv` because the file is in the main folder. Outputs saved to `outputs`.

Method	MAE	WAPE
seasonal_naive	1,429.52	20.95%
weekday_mean	1,243.26	18.22%
timesfm	1,198.04	17.56%

Did not run `evaluate`.

5. User

Choice record: On 10/05/2026, after seeing the validation table,

I chose TimesFM because it had the lowest estimated error on the validation data, with the lowest MAE and WAPE of the three methods.

6. Assistant

****Choice record – October 5, 2026:**** After reviewing the validation table, I chose TimesFM because it had the lowest validation MAE (1,198.04) and WAPE (17.56%) among the three methods.

7. User

run evaluation

8. Assistant

I'll run the evaluation using the same CSV and output folder.

9. User

```
python VD2_analysis.py evaluate --csv data/train.csv --out outputs --choice timesfm
```

10. Assistant

I'll run the evaluation with `--choice timesfm`, using `train.csv` since that is the file's location in your folder.

11. Assistant

Evaluation completed with recorded choice **TimesFM**.

Method	MAE	WAPE
seasonal_naive	1,208.00	17.42%
weekday_mean	1,163.40	16.78%
timesfm	1,161.34	16.75%

TimesFM retained the lowest error, narrowly ahead of weekday_mean. Results saved to `outputs`. Used `train.csv`, the actual file location.

12. User

List the two commands you ran and your Python and package versions (they're in outputs/compare_record.json).

13. Assistant

I'll read `outputs/compare\_record.json` for the recorded versions.

14. Assistant

The two successful commands were:

```
```powershell
python VD2_analysis.py compare --csv train.csv --out outputs
python VD2_analysis.py evaluate --csv train.csv --out outputs --choice timesfm
```
```

Versions recorded in `outputs/compare\_record.json`:

- Python: **3.13.15**
- pandas: **3.0.5**
- NumPy: **2.5.2**

Used `train.csv` because it was in the project folder, rather than a `data` subfolder.

15. User

How many stores and store-days are in the final window for each method? Confirm from `outputs/final_predictions.csv` that no store-date appears twice and no forecast is missing,

16. Assistant

I'll check ``final_predictions.csv`` for each method's store and store-day counts, duplicate store-dates, and missing forecasts.

17. Assistant

Verified directly from ``outputs/final_predictions.csv``. The final window is **June 20–July 31, 2015 (42 days)**.

| Method | Stores | Store-days | Duplicate store-dates | Missing forecasts |
|----------------|--------|------------|-----------------------|-------------------|
| seasonal_naive | 20 | 840 | 0 | 0 |
| weekday_mean | 20 | 840 | 0 | 0 |
| timesfm | 20 | 840 | 0 | 0 |

Each method contains every expected store-date exactly once, with no missing ``prediction`` or ``raw_prediction`` values. The file contains **2,520 rows** total; store-dates repeat across methods as expected, but never within a method.

18. User

can I save this session and come back at a later time?

19. Assistant

I'll check the Codex guidance for returning to this conversation later.

20. Assistant

Yes—you can close this conversation and return later. Reopen the same conversation from your chat/thread history to continue. If you're using the Codex terminal, run ``codex resume`` to select a saved session. [Official OpenAI documentation](https://learn.openai.com/docs/projects)

Your analysis results are already saved in:

``C:\Users\bedna\OneDrive\Documents\AI-Class\VD2_student_files\outputs``

When you return, say: "Continue the VD2 analysis from where we left off."

21. User

i am running in powershell

22. Assistant

Yes. If you launched Codex from PowerShell, exit with ``/exit``. Later, open PowerShell and run:

```
```powershell
codex resume
```

...

Select this session from the list. To resume the most recent session directly:

```
```powershell
codex resume --last
```
```

Your output files are already saved, so you won't need to rerun the analysis.

## 23. User

please check the no peeking rule on store 1. the last training day is June 19th but we have a seasonal naive forecast for june 27. report back your findings

## 24. Assistant

I'll trace Store 1's June 27 seasonal-naive forecast through the code and data to check whether it uses any sales after the June 19 training cutoff.

## 25. Assistant

**Store 1's June 27 seasonal-naive forecast does not violate the no-peeking rule.** It repeats the last observed week, June 13-19, across the entire 42-day forecast window.

| Date    | Actual sales | Seasonal-naive forecast |
|---------|--------------|-------------------------|
| June 13 | 4,256        | Training observation    |
| June 20 | 4,097        | 4,256                   |
| June 27 | 4,019        | 4,256                   |

June 27 uses **June 13's observed sales**, equivalent to repeating June 20's **forecast**. It does not use June 20's actual sales of 4,097.

I also replaced all Store 1 sales after June 19 with dummy values in memory and recalculated: **all 42 seasonal-naive forecasts remained unchanged** and matched the saved predictions. No files were changed.

One qualification: forecasts use future `Open` values under the assignment's explicit assumption that the opening schedule was known at forecast time. Future sales are used for scoring, not generating this forecast.

## 26. User

please show my final accuracy results again

## 27. Assistant

Your final evaluation results cover **20 stores × 42 days = 840 store-days per method**, June 20–July 31, 2015.

| Method                       | MAE ↓           | WAPE ↓        |
|------------------------------|-----------------|---------------|
| Seasonal naive               | 1,208.00        | 17.42%        |
| Weekday mean                 | 1,163.40        | 16.78%        |
| <b>TimesFM – your choice</b> | <b>1,161.34</b> | <b>16.75%</b> |

**\*\*TimesFM had the lowest final error\*\***, narrowly outperforming weekday mean. Lower values are better for both metrics.

**28. User**

please provide the exact absolute error sum and actual sum for the timesfm in the final evaluation? do not calculate WAPE

**29. Assistant**

I'll read the saved TimesFM totals from `final\_scores.csv` without calculating WAPE.

**30. Assistant**

TimesFM's final evaluation totals, exactly as saved in `outputs/final\_scores.csv`:

- **Absolute error sum:** `975525.7464599609`
- **Actual sum:** `5824198.0`

**31. User**

round to 4 decimals

**32. Assistant**

- **Absolute error sum:** `975525.7465`
- **Actual sum:** `5824198.0000`

**33. User**

From outputs/paired\_mae.csv, report two rows, each with its MAE difference and 95% interval:

1, TimesFM minus weekday mean. 2, TimesFM minus Seasonal naive.

**34. Assistant**

I'll read the two TimesFM comparisons from `outputs/paired\_mae.csv`.

**35. Assistant**

From `outputs/paired\_mae.csv`, rounded to four decimals:

| Comparison               | MAE difference | 95% interval         |
|--------------------------|----------------|----------------------|
| TimesFM - weekday mean   | -2.0622        | [-59.2552, 77.6414]  |
| TimesFM - seasonal naive | -46.6622       | [-157.2009, 79.5812] |

Negative differences favor TimesFM. Both intervals include zero.

### 36. User

report MAE for the following: Validation A and Validation B (outputs/validation\_by\_origin.csv), the final window (outputs/final\_scores.csv), and the five Sunday-trading stores versus the other fifteen (outputs/final\_groups.csv)

### 37. Assistant

I'll read the MAE results for all three methods from those output files.

### 38. Assistant

MAE from the requested output files, rounded to four decimals. **\*\*Lower is better.\*\***

| Window / group                    | Seasonal naive       | Weekday mean         | TimesFM              |
|-----------------------------------|----------------------|----------------------|----------------------|
| Validation A: March 28-May 8      | 1763.5643            | <b>**1368.7509**</b> | 1369.0570            |
| Validation B: May 9-June 19       | 1095.4726            | 1117.7600            | <b>**1027.0212**</b> |
| Final: June 20-July 31            | 1208.0024            | 1163.4024            | <b>**1161.3402**</b> |
| Final: five Sunday-trading stores | <b>**1324.1476**</b> | 1500.3810            | 1582.4848            |
| Final: other fifteen stores       | 1169.2873            | 1051.0762            | <b>**1020.9586**</b> |

All dates are in 2015. Bold marks the lowest MAE in each row.

### 39. User

rank highest to lowest

### 40. Assistant

Ranked by **\*\*MAE, highest to lowest\*\*** (worst to best):

| Window / group             | Highest                   | Middle                    | Lowest                    |
|----------------------------|---------------------------|---------------------------|---------------------------|
| Validation A               | Seasonal naive: 1763.5643 | TimesFM: 1369.0570        | Weekday mean: 1368.7509   |
| Validation B               | Weekday mean: 1117.7600   | Seasonal naive: 1095.4726 | TimesFM: 1027.0212        |
| Final window               | Seasonal naive: 1208.0024 | Weekday mean: 1163.4024   | TimesFM: 1161.3402        |
| Five Sunday-trading stores | TimesFM: 1582.4848        | Weekday mean: 1500.3810   | Seasonal naive: 1324.1476 |
| Other fifteen stores       | Seasonal naive: 1169.2873 | Weekday mean: 1051.0762   | TimesFM: 1020.9586        |

### 41. User

please show a comparison of MAE between RAW and KNOWN schedule. do not provide depth analysis just show the compare

### 42. Assistant

I'll read the raw and known-schedule MAEs from `outputs/calendar\_check.csv`.

43. Assistant

| Method         | RAW MAE   | KNOWN schedule MAE |
|----------------|-----------|--------------------|
| Seasonal naive | 1208.0024 | 1208.0024          |
| Weekday mean   | 1163.4024 | 1163.4024          |
| TimesFM        | 1198.0204 | 1161.3402          |

44. User

please show the difference as a column as well

45. Assistant

| Method         | RAW MAE   | KNOWN schedule MAE | Difference (RAW - KNOWN) |
|----------------|-----------|--------------------|--------------------------|
| Seasonal naive | 1208.0024 | 1208.0024          | 0.0000                   |
| Weekday mean   | 1163.4024 | 1163.4024          | 0.0000                   |
| TimesFM        | 1198.0204 | 1161.3402          | 36.6802                  |

46. User

outputs/interval\_coverage.csv, show the TimesFM coverage, row count, and average interval width for all days and for open days only. Do not interpret the results.

47. Assistant

I'll read the all-days and open-days results from `outputs/interval\_coverage.csv`.

48. Assistant

| TimesFM        | Coverage | Row count | Average interval width |
|----------------|----------|-----------|------------------------|
| All days       | 91.0714% | 840       | 4888.3380              |
| Open days only | 90.0000% | 750       | 5474.9385              |

49. User

please create table with the model, validation MAE, Final MAE and Final WAPE. Please leave 2 additional columns empty for me to fill out. 1: Role, 2: Comment

50. Assistant

| Model          | Validation MAE | Final MAE | Final WAPE | Role | Comment |
|----------------|----------------|-----------|------------|------|---------|
| Seasonal naive | 1429.5185      | 1208.0024 | 17.4225%   |      |         |
| Weekday mean   | 1243.2554      | 1163.4024 | 16.7793%   |      |         |
| TimesFM        | 1198.0391      | 1161.3402 | 16.7495%   |      |         |

## 51. User

please create a rendered PDF of this sessions transcription. Include word for word prompts and responses.